

# Learning Python

- `Counter(iterable)` (from `collections`) maps each element to its frequency and is the go-to tool for frequency counting in interview problems.
- `most_common(n)` returns the  $n$  most frequent (`element`, `count`) pairs in descending order (omit  $n$  for all), making it ideal for top- $k$  frequency problems.
- `"".join([char for char in s if char.isalnum()])` strips a string to only letters and digits using `isalnum()`, which returns `True` for a-z, A-Z, and 0-9.
- `char.isnumeric()` accepts any Unicode numeric character (including fractions and superscripts), while `isdigit()` is stricter (ASCII digits 0-9 only).
- Never call `pop(i)` inside a loop that increments `i` forward, as each removal shifts remaining indices; a backward loop such as `for i in range(k, -1, -1): nums.pop(i)` is safe.
- `del nums[i]` is slightly faster than `nums.pop(i)` at the same  $O(n)$  complexity because `pop` has the extra overhead of returning the removed value, which can matter in tight loops over large lists.
- `list.index(i)` returns the index of the first occurrence of value `i` in the list, raising `ValueError` if it is not found.
- `list.count(value)` returns the number of times `value` appears in the list.
- `math.prod(list)` (from `math`) returns the product of all elements in the iterable, returning 1 for an empty iterable.
- `arr.insert(index, new_value)` inserts `new_value` at the given `index`, shifting all subsequent elements one position to the right.

## 1 LeetCode Practice

Table 1: LeetCode practice tracker

| Question                                                        | Level  | Revisit | Done |
|-----------------------------------------------------------------|--------|---------|------|
| <b>1. Arrays &amp; Hashing</b>                                  |        |         |      |
| 217. Contains Duplicate                                         | Easy   |         | ✓    |
| 1. Two Sum                                                      | Easy   |         | ✓    |
| 242. Valid Anagram                                              | Easy   |         | ✓    |
| 383. Ransom Note                                                | Easy   |         | ✓    |
| 349. Intersection of Two Arrays                                 | Easy   |         | ✓    |
| 1512. Number of Good Pairs                                      | Easy   |         | ✓    |
| 205. Isomorphic Strings                                         | Easy   |         | ✓    |
| 49. Group Anagrams                                              | Medium |         | ✓    |
| 347. Top K Frequent Elements                                    | Medium |         | ✓    |
| 238. Product of Array Except Self                               | Medium |         | ✓    |
| <b>2. Two Pointers</b>                                          |        |         |      |
| 344. Reverse String                                             | Easy   |         | ✓    |
| 125. Valid Palindrome                                           | Easy   |         | ✓    |
| 27. Remove Element                                              | Easy   |         | ✓    |
| 26. Remove Duplicates from Sorted Array                         | Easy   |         | ✓    |
| 283. Move Zeroes                                                | Easy   |         | ✓    |
| 977. Squares of a Sorted Array                                  | Easy   |         | ✓    |
| 88. Merge Sorted Array                                          | Easy   |         | ✓    |
| 167. Two Sum II: Input Array Is Sorted                          | Medium |         | ✓    |
| 11. Container With Most Water                                   | Medium | ✓       |      |
| 15. 3Sum                                                        | Medium |         | ✓    |
| <b>3. Sliding Window</b>                                        |        |         |      |
| 643. Maximum Average Subarray I                                 | Easy   |         | ✓    |
| 1876. Substrings of Size Three with Distinct Characters         | Easy   |         | ✓    |
| 2269. Find the K-Beauty of a Number                             | Easy   |         | ✓    |
| 1984. Minimum Difference Between Highest and Lowest of K Scores | Easy   |         | ✓    |
| 1652. Defuse the Bomb                                           | Easy   |         |      |
| 219. Contains Duplicate II                                      | Easy   |         |      |
| 121. Best Time to Buy and Sell Stock                            | Easy   |         |      |
| 3. Longest Substring Without Repeating Characters               | Medium |         |      |
| 567. Permutation in String                                      | Medium |         |      |
| 424. Longest Repeating Character Replacement                    | Medium |         |      |
| <b>4. Stack</b>                                                 |        |         |      |
| 20. Valid Parentheses                                           | Easy   |         |      |
| 682. Baseball Game                                              | Easy   |         |      |
| 1047. Remove All Adjacent Duplicates In String                  | Easy   |         |      |
| 1544. Make The String Great                                     | Easy   |         |      |
| 844. Backspace String Compare                                   | Easy   |         |      |
| 232. Implement Queue using Stacks                               | Easy   |         |      |
| 496. Next Greater Element I                                     | Easy   |         |      |

| Question                                          | Level  | Revisit | Done |
|---------------------------------------------------|--------|---------|------|
| 155. Min Stack                                    | Medium |         |      |
| 150. Evaluate Reverse Polish Notation             | Medium |         |      |
| 739. Daily Temperatures                           | Medium |         |      |
| <b>5. Binary Search</b>                           |        |         |      |
| 704. Binary Search                                | Easy   |         |      |
| 374. Guess Number Higher or Lower                 | Easy   |         |      |
| 35. Search Insert Position                        | Easy   |         |      |
| 278. First Bad Version                            | Easy   |         |      |
| 744. Find Smallest Letter Greater Than Target     | Easy   |         |      |
| 367. Valid Perfect Square                         | Easy   |         |      |
| 69. Sqrt(x)                                       | Easy   |         |      |
| 74. Search a 2D Matrix                            | Medium |         |      |
| 875. Koko Eating Bananas                          | Medium |         |      |
| 153. Find Minimum in Rotated Sorted Array         | Medium |         |      |
| <b>6. Linked List</b>                             |        |         |      |
| 876. Middle of the Linked List                    | Easy   |         |      |
| 83. Remove Duplicates from Sorted List            | Easy   |         |      |
| 203. Remove Linked List Elements                  | Easy   |         |      |
| 206. Reverse Linked List                          | Easy   |         |      |
| 141. Linked List Cycle                            | Easy   |         |      |
| 160. Intersection of Two Linked Lists             | Easy   |         |      |
| 234. Palindrome Linked List                       | Easy   |         |      |
| 19. Remove Nth Node From End of List              | Medium |         |      |
| 2. Add Two Numbers                                | Medium |         |      |
| 143. Reorder List                                 | Medium |         |      |
| <b>7. Trees (DFS and BFS)</b>                     |        |         |      |
| 94. Binary Tree Inorder Traversal                 | Easy   |         |      |
| 104. Maximum Depth of Binary Tree                 | Easy   |         |      |
| 226. Invert Binary Tree                           | Easy   |         |      |
| 100. Same Tree                                    | Easy   |         |      |
| 101. Symmetric Tree                               | Easy   |         |      |
| 112. Path Sum                                     | Easy   |         |      |
| 543. Diameter of Binary Tree                      | Easy   |         |      |
| 102. Binary Tree Level Order Traversal            | Medium |         |      |
| 98. Validate Binary Search Tree                   | Medium |         |      |
| 230. Kth Smallest Element in a BST                | Medium |         |      |
| <b>8. Heap / Priority Queue</b>                   |        |         |      |
| 1464. Maximum Product of Two Elements in an Array | Easy   |         |      |
| 1046. Last Stone Weight                           | Easy   |         |      |
| 2558. Take Gifts From the Richest Pile            | Easy   |         |      |
| 506. Relative Ranks                               | Easy   |         |      |
| 1337. The K Weakest Rows in a Matrix              | Easy   |         |      |
| 2335. Minimum Amount of Time to Fill Cups         | Easy   |         |      |
| 703. Kth Largest Element in a Stream              | Easy   |         |      |
| 215. Kth Largest Element in an Array              | Medium |         |      |

| Question                            | Level  | Revisit | Done |
|-------------------------------------|--------|---------|------|
| 973. K Closest Points to Origin     | Medium |         |      |
| 621. Task Scheduler                 | Medium |         |      |
| <b>9. Backtracking</b>              |        |         |      |
| 509. Fibonacci Number               | Easy   |         |      |
| 231. Power of Two                   | Easy   |         |      |
| 326. Power of Three                 | Easy   |         |      |
| 590. N-ary Tree Postorder Traversal | Easy   |         |      |
| 257. Binary Tree Paths              | Easy   |         |      |
| 1863. Sum of All Subset XOR Totals  | Easy   |         |      |
| 401. Binary Watch                   | Easy   |         |      |
| 78. Subsets                         | Medium |         |      |
| 46. Permutations                    | Medium |         |      |
| 39. Combination Sum                 | Medium |         |      |
| <b>10. Graphs</b>                   |        |         |      |
| 1791. Find Center of Star Graph     | Easy   |         |      |
| 997. Find the Town Judge            | Easy   |         |      |
| 1436. Destination City              | Easy   |         |      |
| 733. Flood Fill                     | Easy   |         |      |
| 463. Island Perimeter               | Easy   |         |      |
| 1971. Find if Path Exists in Graph  | Easy   |         |      |
| 993. Cousins in Binary Tree         | Easy   |         |      |
| 200. Number of Islands              | Medium |         |      |
| 994. Rotting Oranges                | Medium |         |      |
| 207. Course Schedule                | Medium |         |      |
| <b>11. Dynamic Programming</b>      |        |         |      |
| 70. Climbing Stairs                 | Easy   |         |      |
| 1137. N-th Tribonacci Number        | Easy   |         |      |
| 746. Min Cost Climbing Stairs       | Easy   |         |      |
| 118. Pascal's Triangle              | Easy   |         |      |
| 119. Pascal's Triangle II           | Easy   |         |      |
| 338. Counting Bits                  | Easy   |         |      |
| 392. Is Subsequence                 | Easy   |         |      |
| 198. House Robber                   | Medium |         |      |
| 322. Coin Change                    | Medium |         |      |
| 300. Longest Increasing Subsequence | Medium |         |      |